

Practical NoSQL Database

CHAPTER 2

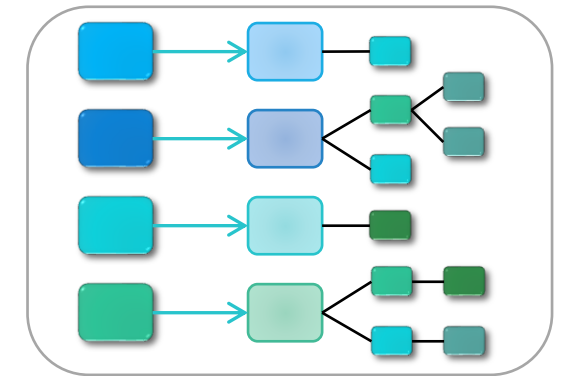
DATA MODELLING AND RELATIONSHIP IN NoSQL

Outline

- ❖ Data Modeling concept
- ❖ Relationships

Document Structure

- ❖ structure of documents
- ❖ Relationships between data



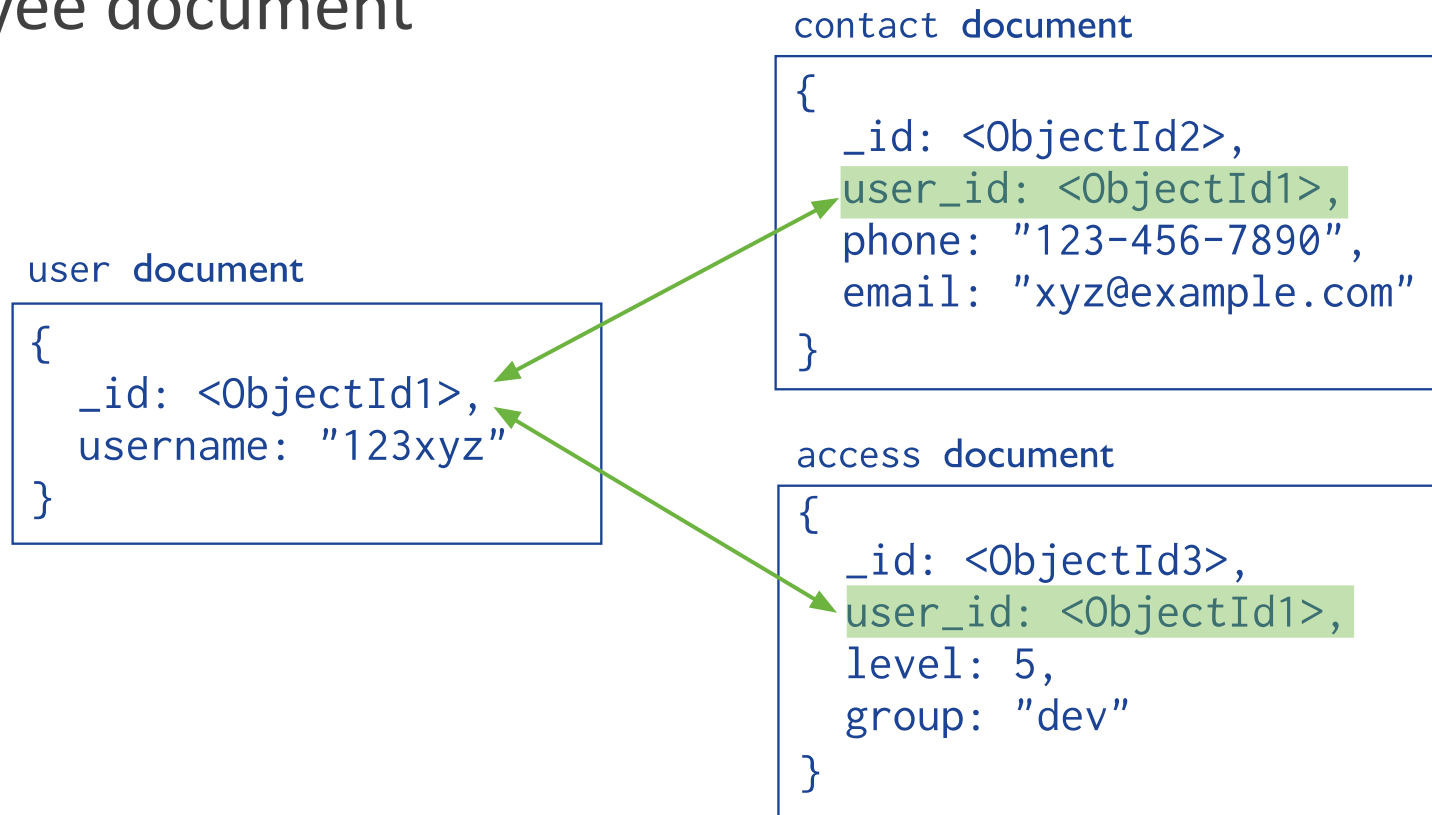
Document Store

Document Structure

- ❖ Tools to represent the relationships:
 - ❖ References
 - ❖ Embedded Data

Data Modeling using References

❖ Employee document



Data Modeling using Embedded Data

```
{
  _id: <ObjectId>,
  username: "123xyz",
  contact: {
    phone: "123-456-7890",
    email: "xyz@example.com"
  },
  access: {
    level: 5,
    group: "dev"
  }
}
```

Embedded sub-document

Embedded sub-document

Relationships in NoSQL

- ❖ One-To-One (1:1) relationship
- ❖ One-To-Many (1:M) relationship
- ❖ Many –To-Many (M:N) relationship

On—To-One Relationship

Referencing

```
{
  _id: "joe",
  name: "Joe Bookreader"
}

{
  patron_id: "joe",
  street: "123 Fake Street",
  city: "Faketon",
  state: "MA",
  zip: "12345"
}
```

Embedding

```
{
  _id: "joe",
  name: "Joe Bookreader",
  address: {
    street: "123 Fake Street",
    city: "Faketon",
    state: "MA",
    zip: "12345"
  }
}
```

One-To-Many Relationship

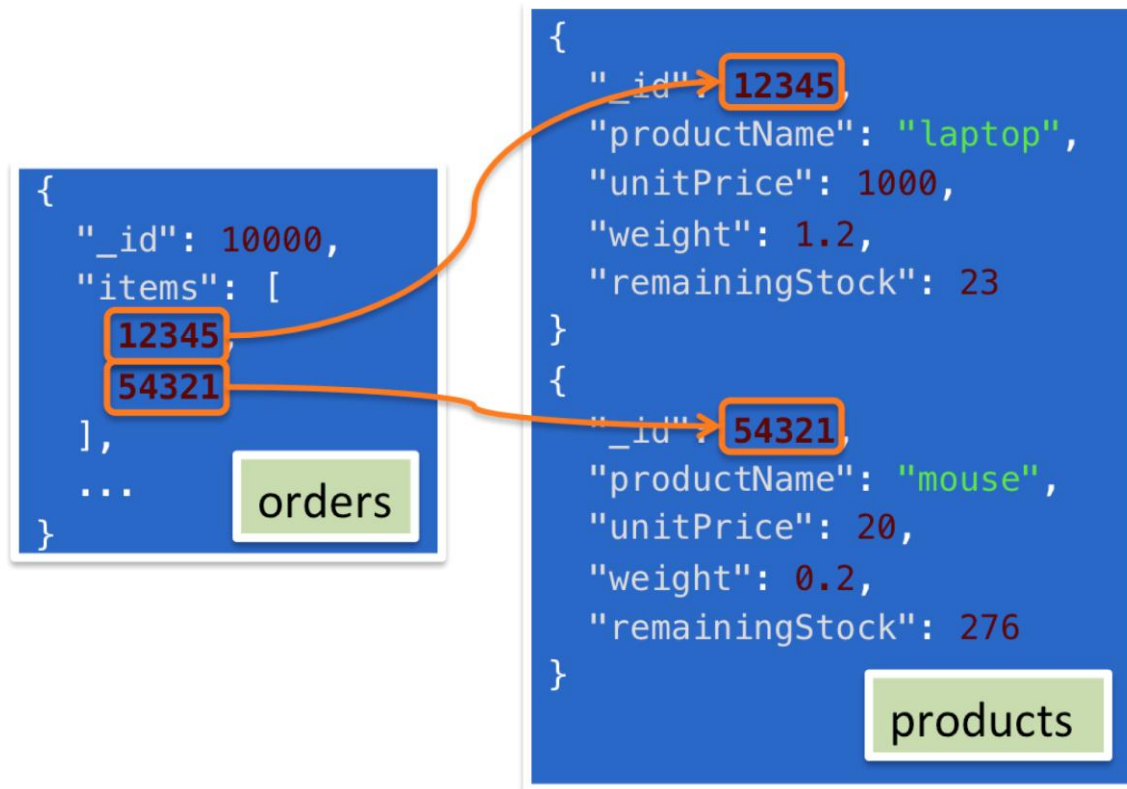
```
{
  _id: "joe",
  name: "Joe Bookreader"
}
{
  patron_id: "joe",
  street: "123 Fake Street",
  city: "Faketon",
  state: "MA",
  zip: "12345"
}
{
  patron_id: "joe",
  street: "1 Some Other Street",
  city: "Boston",
  state: "MA",
  zip: "12345"
}
```

Referencing

```
{
  _id: "joe",
  name: "Joe Bookreader",
  addresses: [
    {
      street: "123 Fake Street",
      city: "Faketon",
      state: "MA",
      zip: "12345"
    },
    {
      street: "1 Some Other Street",
      city: "Boston",
      state: "MA",
      zip: "12345"
    }
  ]
}
```

Embedding

Many-To-Many Relationship



Two Way embedding

Document Format

Column name

Values

```
" Title " : " CouchDB ",
" Last editor " : "172.5.123.91" ,
" Last modified " : "9/23/2010" ,
" Categories " : [" Database ", " NoSQL ", " Document Database "],
" Body " : " CouchDB is a ..." ,
" Reviewed " : false
```

Create Collection and Drop

```
db.createCollection (< collection >);
```

```
db. < collection >. Drop();
```

Insert / find / save command

Table name

Operation

```
db.< collection >. insert ( { title : " MongoDB ", last_editor : ... } );
```

```
db.< collection >. find ( { categories : [ " NoSQL ", " Document  
Databases " ] } );  
db.< collection >. save ( { ... } );
```

Summary

- ❖ Data Model concept
- ❖ Relationship
- ❖ NoSQL command : Create database/ Collection / insert/ find / save